ST2334 Cheatsheet

AY24/25 - Anselm Long

1 Basic Concepts of Probability

1.1 Event Operations and Relationships

Let A and B be two events in the sample space. Some event operations include:

- Union - $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- Intersection - $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- Complement - $A' = \{x : x \in S \text{ but } x \notin A\}$

Event Relationships:

- Contained - $A \subset B$
- Mutually Exclusive - $A \cap B = \emptyset$
- Equivalent - $A \subset B$ and $B \subset A$, then $A = B$

$A \cup B = A \implies B \subset A$
$A \cap B = A \implies A \subset B$

More Event Operations:

- $A \cap A' = \emptyset$
- $A \cup A' = S$
- $A \cap \emptyset = \emptyset$
- $(A')' = A$ (Complement)
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributive)
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive)
- $A \cup B = A \cup (B \cap A')$ (Distributive with Complement)
- $A = (A \cap B) \cup (A \cap B')$ (Disjoint Subsets)
- $(A \cap B)' = A' \cup B', \quad (A \cup B)' = A' \cap B'$ (De Morgan's Law)

1.2 Counting Methods

Multiplication Principle - if r different experiments are performed sequentially, and each experiment results in an outcome n, total outcomes are derived by multiplying all n together.

Addition Principle - if an experiment can be performed k different ways that *do not overlap*, the total number of ways we can perform the experiment is $n_1 + n_2 + \cdots + n_k$

Permutation - a *selection* and *arrangement* of r objects out of n. *Order is considered.*

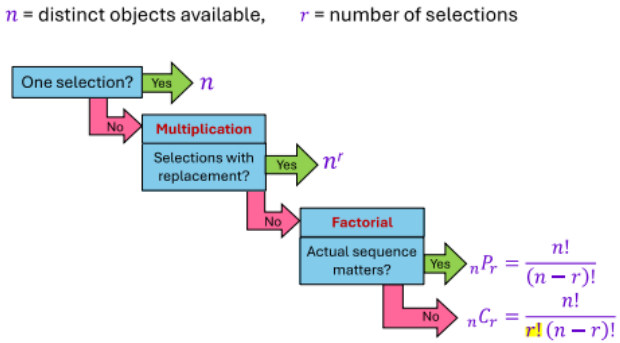
$$P_r^n = \frac{n!}{(n-r)!}$$

When $r = n$, $P_n^n = n!$. This is the number of ways to arrange n objects in order. Phrases to look out for: *how many ways*. Use permutation over combination if the objects are named.

Combination - a *selection* of r objects out of n, without regard to the order.

$$C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}$$

Permutation can be decided as choosing r objects out of n first, then permutating r objects - $\binom{n}{r} \times P_r^r$. Look out for: *number of … to select, unnamed objects*



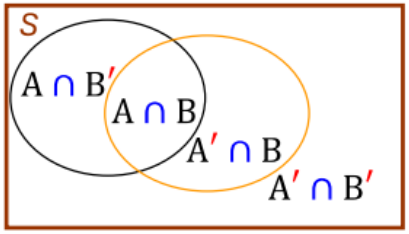
Axioms of Probability

1. For any event A, $0 \le P(A) \le 1$
2. For the sample space, $P(S) = 1$
3. Two events A and B are mutually exclusive $\iff A \cap B = \emptyset, \implies P(A \cup B) = P(A) + P(B)$

Properties of Probability

1. Probability of the empty set $\emptyset$ is $P(\emptyset) = 0$
2. For mutually exclusive events, we have $P(A_1 \cup A_2 \cup \cdots \cup A_n) = P(A_1) + P(A_2) + \cdots + P(A_n)$
3. For any event A, we have $P(A') = 1 - P(A)$ (Complement)
4. For any two events A and B, $P(A) = P(A \cap B) + P(A \cap B')$ (Splitting up into complements)
5. For any two events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (Inclusion-exclusion Principle)
6. If $A \subset B$, then $P(A) \le P(B)$.

$$S = (A \cap B') \cup (A \cap B) \cup (A' \cap B) \cup (A' \cap B')$$



$$A \cup B = A \cup (A' \cap B)$$
$$A \cup B = (A \cap B') \cup (A \cap B) \cup (A' \cap B)$$
$$A \cup B = (A \cap B') \cup B$$
$$A \cup B = (A' \cap B')'$$

1.3 Conditional Probability

Definition

For any two events A and B with $P(A) > 0$, the conditional probability of B given that A has occurred is defined by

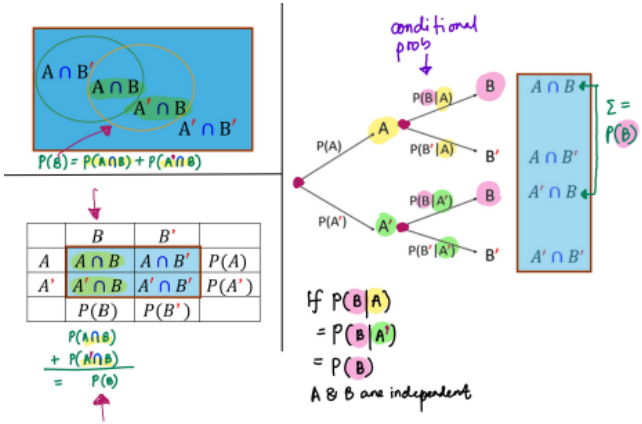
$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

We regard A is our *reduced sample space*.
Then, the **Multiplication Rule** is defined as:

$$P(A \cap B) = P(A)P(B|A), \text{ if } P(A) \neq 0 \text{ or } P(B)P(A|B), \text{ if } P(B) \neq 0$$

Inverse Probability Formula

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$



1.4 Independence

Learning that the event B occurred gives us *no information* that would change our probabilities for another event A occurring. Two events A and B are **independent** *if and only if*

$$P(A \cap B) = P(A)P(B), \text{ denoted by } A \perp B$$

Then, if $P(A) \neq 0$, $A \perp B \iff P(B|A) = P(B)$, as the knowledge of A does not change the probability of B.

	If A & B are disjoint /mutually exclusive ($A \cap B = \phi$)	If A & B are independent ($A \perp B$) (A, B totally unrelated)
$P(A \cup B)$	$P(A) + P(B)$	$P(A) + P(B) - P(A)P(B)$
$P(A \cap B)$	0	$P(A)P(B)$
$P(A \mid B)$	0	$P(A \mid \cancel{B}) = P(A)$
$P(B \mid A)$	0	$P(B \mid \cancel{A}) = P(B)$

Useful Results:

- For events A and B where $P(A) > 0, P(B) > 0$, if $A \perp B$, A and B are not mutually exclusive as $P(A \cap B) = P(A)P(B) > 0$.
- Similarly, if A and B are mutually exclusive, then $A \not\perp B$ as $P(A \cap B) = 0 \neq P(A)P(B)$
- S and $\emptyset$ are independent of any other event.
- If $A \perp B$, then $A \perp B', A' \perp B,$ and $A' \perp B'$.

1.5 The Law of Total Probability

Definition For any events A and B, we have

$$P(B) = P(A)P(B|A) + P(A')P(B|A')$$

Splitting the probability of B into cases where A happened and where A didn't happen. Compute this with a *probability tree!*

1.6 Bayes' Theorem

Definition

$$P(A_k|B) = \frac{P(A_k)P(B|A_k)}{\sum_{i=1}^{n} P(A_i)P(B|A_k)}$$

Intuitively, probability of a certain partition of S, A_k, happening, given B happens – is splitting up $P(A \cap B)$ into conditional probability, then dividing over the reduced sample space $P(B)$ – which is a sum of the partitions.

Special Case when $n = 2$:

$$P(A|B) = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A')P(B|A')} = \frac{P(A)P(B|A)}{P(B)}$$

Note that $P(A|B) = 1 - P(A'|B)$, due to the reduced sample space.

Useful Results

- "At least n...": Count total number of ways - ways with nothing
- No items are allowed to be grouped together, eg. women don't want to sit together: seat and arrange one party first, select r out of n slots and permutate the r items.
- Let S be a sample space and A be an event. If $x \in S$ but $x \notin A$, $P(A) < 1$ does not always hold, in the case of continuous random variable, $P(X = x) = 0$ so $P(A)$ can still be 1.

Integrate Exponential Functions

$$\int e^x dx = e^x + C$$

$$\int ae^{ax} dx = e^{ax} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int a^x \ln a dx = a^x + C$$

2 Random Variables - pdf, cdf, E(X), V(X)

2.1 Probability Distribution

Discrete Random Variable (DRV) - number of values in R_x is finite or countable

Probability Mass Function of a DRV:
Important to prove that something is a probability mass function!

1. $f(x_i) \geq 0$ for all $x_i \in R_X$
2. $f(x) = 0$ for all $x \notin R_X$
3. $\sum_{i=1}^{\infty} f(x_i) = 1$
4. For any set $B \subset \mathbb{R}$ we have $P(X \in B) = \sum_{x_i \in B \cap R_X} f(x_i)$ (add probabilities where x is in B)

Continuous Random Variable - R_X is an interval or a collection of intervals

Probability Density Function of a CRV:

1. $f(x) \geq 0$ for all $x \in R_X$
2. $f(x) = 0$ for $x \notin R_X$
3. $\int_{R_X} f(x)dx = 1$
4. For any a and b such that $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(x)dx$

Remarks:

- $\int_{-\infty}^{\infty} f(x)dx = 1$
- $P(X = x_0) = \int_{x_0}^{x_0} f(x)dx = 0$. $P(A) = 0$, but A is not $\emptyset$
- $P(a < X < b) = P(a < X \leq b) = P(a \leq X < b) = P(a \leq X \leq b)$

To check if a function is a pdf, check that condition 1, 2, and 3 hold.

2.2 Cumulative Distribution Function (cdf)

For any random variable X, we define its cdf by

$$F(x) = P(X \leq x)$$

CDF for DRV:
The value of $F(x)$ is the sum of the values of $f(t)$ up to x.

$$F(x) = \sum_{t \in R_X; t \leq x} f(t) = \sum_{t \in R_X; t \leq x} P(X = t) \tag{1}$$

For 2 numbers $a < b$, we have

$$P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a-)$$

CDF for CRV:
Value of $F(x)$ is the integral of $f(t)$.

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

Then,

$$f(x) = \frac{dF(x)}{dx}$$

Further, $P(a \leq X \leq b) = P(a < X < b) = F(b) - F(a)$
$F(x)$ is *non-decreasing*, $x_1 < x_2, F(x_1) \leq F(x_2)$
Range of values of $F(x)$ and $f(x)$:

- $0 \leq F(x) \leq 1$
- For discrete, $0 \leq f(x) \leq 1$
- For continuous, $f(x) \geq 0$ but not necessary that $f(x) \leq 1$ as integral over space is $= 1$

Note that for a cumulative distribution function, the comparison signs have to be less than equal, not just less than.

2.3 Expectation and Variance

Expectation - DRV

$$\mu_X = E(X) = \sum_{x_i \in R_x} x_i f(x_i)$$

Basically take x multiplied by $f(x)$ for every x in the function.

Expectation - CRV

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x f(x)dx$$

Integrate $xf(x)$. Note that the mean of X is not necessarily a possible value of the random variable X.

Properties of Expectation

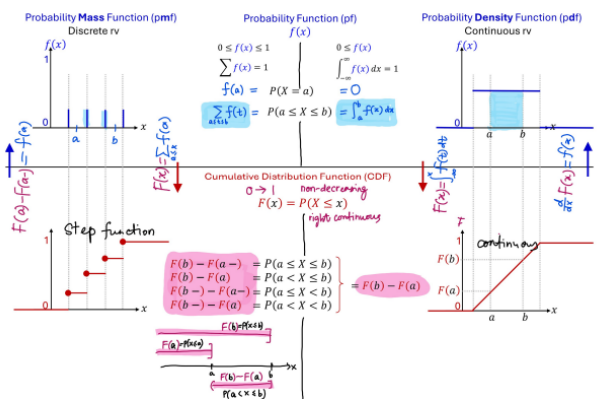
1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$

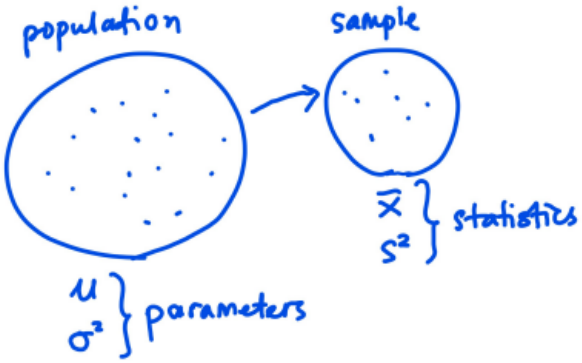
Variance

$$\sigma_X^2 = V(X) = E(X - \mu_X)^2$$
$$V(X) = E(X^2) - [E(X)]^2$$

Remarks

- Holds for both discrete and continuous.
- Discrete - $V(X) = \sum_{x \in R_X} (x - \mu_X)^2 f(x)$
- Continuous - $V(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx$
- $V(X) \geq 0$. $V(X) = 0 \iff P(X = E(X)) = 1$ i.e. X is constant
- $V(aX + b) = a^2 V(X)$
- $\sigma_X = \sqrt{V(X)}$
- Variance of a constant $= 0$. Eg. $V(E(X)) = 0$





Which of the following is the probability function of X?

$$F_X(x) = \begin{cases} 0, & x < 1; \\ 0.3, & 1 \leq x < 3; \\ 0.4, & 3 \leq x < 4; \\ 0.45, & 4 \leq x < 6; \\ 0.6, & 6 \leq x < 12; \\ 1, & 12 \leq x. \end{cases}$$

(a)

x	1	3	4	6	12
$f_X(x)$	0.3	0.1	0.05	0.15	0.4

(b)

x	1	3	4	6	12
$f_X(x)$	0.3	0.1	~~0.45~~	0.05	0.4

(c)

x	1	3	4	6	12
$f_X(x)$	~~0~~	0.3	0.4	0.6	1

(d) None of the given options.

Discrete X	rv X	Continuous X
$0 \leq f(x) \leq 1$	prob func, $f(x)$	$0 \leq f(x)$
$\sum f(x) = 1$	total prob $= 1$	$\int_{-\infty}^{\infty} f(x)\,dx = 1$
$\sum x f(x)$	$E(X)$	$\int_{-\infty}^{\infty} x f(x)\,dx$
$\sum x^2 f(x)$	$E(X^2)$	$\int_{-\infty}^{\infty} x^2 f(x)\,dx$

$V(X) = E(X^2) - [E(X)]^2$

two rv (X & Y)

$0 \leq f(x,y) \leq 1$ ← Joint prob func $f(x,y)$ → $0 \leq f(x,y)$

$\sum\sum f(x,y) = 1$ ← total prob $= 1$ → $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

$\sum\sum (xy) f(x,y)$ ← $E(XY)$ → $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (xy) f(x,y)\,dx\,dy$

$\mathrm{cov}(XY) = E(XY) - E(X)E(Y)$

3 Joint Distributions - Marginal, Conditional, Covariance

3.1 Joint Distributions

(X, Y) is a two-dimensional random vector.
Range Space of (X, Y):

$$R_{X,Y} = \{(x, y) \mid x = X(s), y = Y(s), s \in S\}.$$

Discrete Joint Probability Function

$$f_{X,Y}(x, y) = P(X = x, Y = y), \quad (x, y) \in R_{X,Y}$$

Properties:

1. $f_{X,Y}(x, y) \geq 0$ for any $(x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0$ for any $(x, y) \notin R_{X,Y}$
3. $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_{X,Y}(x_i, y_j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = 1$
4. Let A be a subset of $R_{X,Y}$, then $P((X, Y) \in A) = \sum\sum_{(x,y)\in A} f_{X,Y}(x, y)$

Note that $P(X \leq x, Y \leq y) \neq 1 - P(X > x, Y > y)$

Continuous Joint Probability Function

$$P((X, Y) \in D) = \iint_{(x,y)\in D} f_{X,Y}(x, y)\ dy\ dx$$

$$P(a \leq X \leq b, c \leq X \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y)\ dy\ dx$$

Properties:

1. $f_{X,Y}(x, y) \geq 0$, for any $(x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0$, for any $(x, y) \notin R_{X,Y}$
3. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\ dx\ dy = 1$

3.2 Marginal and Conditional Distributions

Marginal distribution of X is the individual distribution of X, ignoring the value of Y.
Discrete:

$$f_X(x) = \sum_y f_{X,Y}(x, y)$$

Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\ dy$$

$f_X(x)$ is a probability function, so it satisfies all properties of it.
Conditional Distribution of Y given $X = x$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Remarks:

- Remember to calculate joint probability to find conditional!
- $f_{Y|X}(y|x)$ is a probability function, so it satisfies all the properties of a probability function. However, it is not a probability function for x, so integral or summation of the function does not equal to 1.
- If $f_X(x) > 0$, $f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y|x)$, and vice versa.
- We can compute $P(Y \leq y|X = x) = \int_{-\infty}^{y} f_{Y|X}(y|x)\ dy$ and then $E(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)\ dy$. This can be established for the discrete case similarly.

$f(x,y) \leq f(x)$ if X is discrete

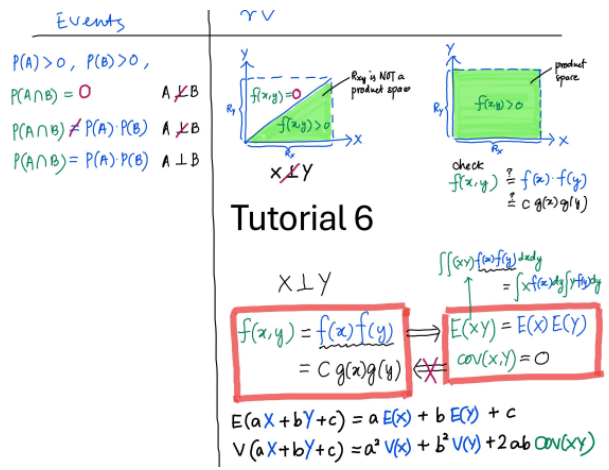
$\int_x\int_y f(x,y)\,dx\,dy \leq \int_x f(x)\,dx$ if X is continuous

3.3 Independent Random Variables

Random variables X and Y are **independent** if and only if for any x and y,

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

This applies whether it is continuous or discrete. If $R_{X,Y}$ is not a product space, then X and Y are *not independent*.



Properties:
For independent random variables X and Y,

1. $P(X \leq x; Y \leq y) = P(X \leq x)P(Y \leq y)$
2. For arbitrary functions $g_1(\cdot)$ and $g_2(\cdot)$, $g_1(X)$ and $g_2(Y)$ are independent, eg. $sin(X)$ and $cos(Y)$
3. Independence is connected with conditional distribution. If $f_X(x) > 0$, then $f_{Y|X}(y|x) = f_Y(y)$, and vice versa for y.

Checking Independence

- $R_{X,Y}$ is a product space where x and y can be factorised.
- $f_{X,Y}(x, y) = C \times g_1(x) \times g_2(y)$ eg. $y(x^2 + x)$

Marginal Distribution under Independence
We can derive the probability distributions for X and Y by standardizing $g_1(x)$ and $g_2(y)$. For the discrete case:

$$f_X(x) = \frac{g_1(x)}{\sum_{t\in R_X} g_1(t)}$$

For continuous case:

$$f_X(x) = \frac{g_1(x)}{\int_{t\in R_X} g_1(t)\,dt}$$

3.4 Expectation and Covariance

For the discrete case, expectation of a random vector is:

$$E(g(X, Y)) = \sum_x \sum_y g(x, y) f_{X,Y}(x, y)$$

For continuous case:

$$E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y)\,dy\,dx$$

Covariance

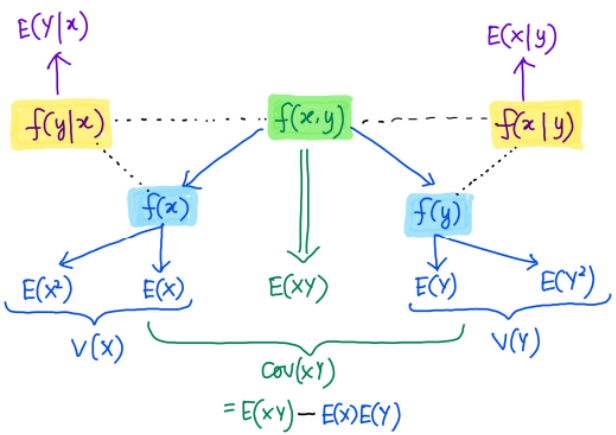
$$cov(X, Y) = E[(X - E(X))(Y - E(Y))]$$

Calculation
Discrete - $\text{cov}(X, Y) = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f_{X,Y}(x, y)$
Continuous -
$\text{cov}(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{X,Y}(x, y)\,dx\,dy$
Properties

1. $cov(X, Y) = E(XY) - E(X)E(Y)$
2. If X and Y are independent, then $\text{cov}(X, Y) = 0$. $\text{cov}(X, Y) = 0$ does not imply that they are independent.
3. This implies that if X and Y are independent, $E(XY) = E(X)E(Y)$
4. $cov(aX + b, cY + d) = ac \cdot cov(X, Y)$
   - $cov(X, Y) = cov(Y, X)$
   - $cov(X + b, Y) = cov(X, Y)$
   - $cov(aX, Y) = a\text{cov}(X, Y)$
5. $V(aX + bY) = a^2V(X) + b^2V(Y) + 2ab \cdot cov(X, Y)$
   - $V(aX) = a^2V(X)$
   - $V(X + Y) = V(X) + V(Y) + 2cov(X, Y)$
   - For X and Y that are independent, $V(X \pm Y) = V(X) + V(Y)$
   - For random variables $X_1, X_2, \ldots, X_n$, we have:

$$V(X_1+X_2+\cdots+X_n) = V(X_1)+V(X_2)+\cdots+V(X_n)+2\sum_{j>i} \text{cov}(X_i, X_j)$$

We have to add twice the covariance between pairs of random variables.



		X: x_1	x_2	row total
Y	y_1	$f(x_1, y_1)$	$f(x_2, y_1)$	$\sum f(x_i, y_1) = f(y_1)$
	y_2	$f(x_1, y_2)$	$f(x_2, y_2)$	$\sum f(x_i, y_2) = f(y_2)$
	y_3	$f(x_1, y_3)$	$f(x_2, y_3)$	$\sum f(x_i, y_3) = f(y_3)$
col total		$\sum f(x_1, y_j) = f(x_1)$	$\sum f(x_2, y_j) = f(x_2)$	$\sum\sum f(x_i, y_j) = 1$

$\sum f(y_j) = 1$

$\sum f(x_i) = 1$

marginal prob function

$$\sum_{\text{all } y} f(x,y) = f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

$$\sum_{\text{all } x} f(x,y) = f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

$$f(x|y) = \frac{f(x,y)}{f(y)}$$

$$E(X|y) = \int_{-\infty}^{\infty} (x) f(x|y)\,dx$$

$$f(y|x) = \frac{f(x,y)}{f(x)}$$

$$E(Y|x) = \int_{-\infty}^{\infty} (y) f(y|x)\,dy$$

If $X \perp Y$,

$$f(x_i, y_j) = f(x_i) \times f(y_j)$$

$$E(XY) = E(X) \times E(Y)$$

$$E(X|y) = E(X)$$

$$E(Y|x) = E(Y)$$

$$\sum_{\text{all } x} f(x,y) = f(y)$$

$$0 \le f(y) \le 1$$

$$\int_{-\infty}^{\infty} f(x,y)\,dx = f(y)$$

$$0 \le f(y) \le 1 \text{ ?}$$

4 Probability Distributions - Binom, NB, Exp, Poisson, Normal etc.

4.1 Discrete Distributions

4.1.1 Discrete Uniform Distribution

$$f_X(x) = \begin{cases} \frac{1}{k}, & x = x_1, x_2, \ldots, x_k; \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

$$\mu_X = E(X) = \frac{1}{k}\sum_{i=1}^{k} x_i$$

$$V(X) = \frac{1}{k}\sum_{i=1}^{k} x_i^2 - \mu_X^2$$

4.1.2 Bernoulli Trial

A random experiment with only 2 possible outcomes.
Bernoulli Random Variable - has only two possible values, 1 or 0

$$f_X(x) = P(X = x) = p^x(1-p)^{1-x} \text{ for x = 0, 1}$$

For a Bernoulli Random Variable, we have $\mu_X = E(X) = p$, and $\sigma_X^2 = V(X) = p(1-p) = pq$
Bernoulli Process - sequence of repeatedly performed independent and identical Bernoulli trials

4.1.3 Binomial Distribution

Binomial Random Variable counts the number of successes in n trials of a Bernoulli Process. Where the probability of success for each trial is the same p, and the trials are independent. Then, $X \sim \text{Bin}(n, p)$
The probability of getting exactly x successes is:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} \text{ for x = 0, 1, 2, ..., n}$$

Then, $E(X) = np$, and $V(X) = np(1-p)$

4.1.4 Negative Binomial Distribution

Let X be the number of independent and identically distributed Bernoulli(p) trials needed until the kth success occurs. Then X follows a Negative Binomial Distribution, denoted by $X \sim NB(k, p)$

$$f_X(x) = P(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \text{for x = k, k + 1, k +2 ...}$$

Then, $E(X) = \frac{k}{p}$ and $V(X) = \frac{(1-p)k}{p^2}$

4.1.5 Geometric Distribution

Let X be the number of i.i.d Bernoilli(p) trials needed until the *first* success occurs. Then X follows a **Geometric Distribution**, denoted by $X \sim \text{Geom}(p)$. Basically equivalent to $X \sim NB(1, p)$.
Probability mass function of X:

$$f_X(x) = P(X = x) = (1-p)^{x-1}p$$

$E(X) = \frac{1}{p}$, $V(X) = \frac{1-p}{p^2}$

4.1.6 Poisson Distribution

The Poisson random variable X denotes the *number of events occuring* in a fixed period of time or fixed region. We denote $X \sim \text{Poisson}(\lambda)$, where the parameter $\lambda > 0$ is the expected number of occurrences during the given period.

$$f_X(k) = P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

where k = 0, 1, ... is the number of occurrences of such events. Then, $E(X) = \lambda$ and $V(X) = \lambda$.
To calculate $P(X > k)$, we take

$$1 - P(X \le k) = 1 - \sum_{x=0}^{k} \frac{e^{-\lambda}\lambda^x}{x!}$$

Poisson Process Count the number of occurrences within some interval of time. For a rate parameter α :

- the expected number of occurrences in an interval of length T is αT
- there are no simultaneous occurrences
- the number of occurrences in disjoint time intervals are independent
- number of occurences in any interval T of a Poisson process follows a Poisson(αT) distribution.

Binomial Approximation Where $n \to \infty$ and $p \to 0$ so that $\lambda = np$ remains constant, $X \sim \text{Poisson}(np)$. This holds when $n \ge 20$, $p \le 0.05$ or if $n \ge 100$, $np \le 10$.

4.2 Continuous Distributions

4.2.1 Continuous Uniform Distribution

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

Denoted by $X \sim U(a, b)$, $E(X) = \frac{a+b}{2}$, $V(X) = \frac{(b-a)^2}{12}$
Cumulative Distribution Function:

$$F_X(x) = \begin{cases} 0, & x < a; \\ \frac{x-a}{b-a}, & a \le x \le b; \\ 1, & x > b \end{cases}$$

Exponential Distribution Models the waiting time to the first success in continuous time. $X \sim \text{Exp}(\lambda)$. λ here does not denote the mean.

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \ge 0 \\ 0, & x < 0 \end{cases}$$

$$E(X) = \frac{1}{\lambda},\ V(X) = \frac{1}{\lambda^2}$$

$$F_X(x) = P(X \le x) = \begin{cases} 1 - e^{-\lambda x}, & x \ge 0 \\ 0, & x < 0 \end{cases}$$

Then, $P(X > x) = e^{-\lambda x}$, for $x > 0$. Alternatively, we let $\mu = \frac{1}{\lambda}$, then

$$f_X(x) = \begin{cases} \frac{1}{\mu}e^{-x/\mu}, & x \ge 0; \\ 0, & x < 0. \end{cases} \tag{3}$$

$E(X) = \mu,\ V(X) = \mu^2$
Memoryless Property of Exponential Distribution

$$P(X > s + t | X > s) = P(X > t)$$

4.2.2 Normal Distribution

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \quad -\infty < x < \infty. \tag{4}$$

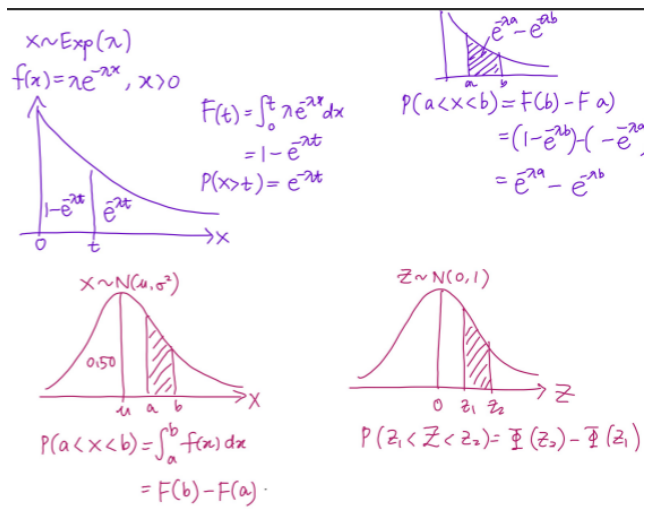
We denote $X \sim N(\mu, \sigma^2)$.
It can be shown that $E(X) = \mu$ *and* $V(X) = \sigma^2$.
Properties

- Total area under the curve = 1
- Mean changes position of the curve
- Standard deviation adjusts how flat or sharp the curve is
- If we let $Z = \frac{X-\mu}{\sigma}$, then $Z \sim N(0, 1)$. This is the standard normal distribution.
- We calculate normal probability using the standard normal

Standard Normal Distribution

$$P(x_1 < X < x_2) = P(z_1 < Z < z_2) = \Phi(\frac{x_2 - \mu}{\sigma}) - \Phi(\frac{x_1 - \mu}{\sigma})$$

For any z, $\Phi(z) = P(Z \le z) = P(Z \ge -z) = 1 - \Phi(-z)$



Quartile The number x_α that satisfies $P(X \ge x_\alpha) = \alpha$.
Normal Approximation to Binomial Let $X \sim Bin(n, p)$. When $n \to \infty$ but p remains a constant. Used when $np > 5$ and $n(1-p) > 5$. Then,

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} = \frac{X - np}{\sqrt{np(1-p)}} \text{ is approximately } \sim N(0, 1). \tag{5}$$

But, we have to employ **continuity correction** here.
Continuity Correction is using a continuous distribution (normal) to find a discrete distribution like binomial. This is used when n is large.

$$P(X = k) \approx P(k - \frac{1}{2} < X < k + \frac{1}{2})$$

$$P(a \le X \le b) \approx P(a - \tfrac{1}{2} < X < b + \tfrac{1}{2})$$

$$P(a < X \le b) \approx P(a + \tfrac{1}{2} < X < b + \tfrac{1}{2})$$

$$P(a \le X < b) \approx P(a - \tfrac{1}{2} < X < b - \tfrac{1}{2})$$

$$P(a < X < b) \approx P(a + \tfrac{1}{2} < X < b - \tfrac{1}{2})$$

$$P(X \le c) = P(0 \le X \le c) \approx P(-\tfrac{1}{2} < X < c + \tfrac{1}{2})$$

$$P(X > c) = P(c < X \le n) \approx P(c + \tfrac{1}{2} < X < n + \tfrac{1}{2})$$

Discrete Uniform(k)
Bernoulli(p)
Binomial(n,p)
Hypergeometric(N, S, n)
Poisson(λ)
Geometric(p)
Negative Binomial(k,p)

Continuous Uniform(a, b)
Exponential($\lambda = \frac{1}{\mu}$)
Normal(μ, σ^2)

Tutorial 7

rv $X \sim$ Distribution name(parameter)

$f(x)$ $P(X=x)$ $P(X \le x)$ $P(X > x)$

$E(X)$

$V(X)$

5 Sampling and Sampling Distributions - CLT

5.1 Population and Sample

The totality of all possible outcomes or observations of a survey or experiment is called a **population.** A **sample** is any subset of a population.

5.2 Random Sampling

A population can be modeled using a distribution. But we don't know the values of the parameters. We then use a sample to learn about the parameters. A **simple random sample** is a sample chosen such that every subset of n observations of the population has the same probability of being selected.

Sampling Distribution of Sample Mean Suppose a random sample of n observations $(X_1, ...X_n)$ has been taken. A function of that is calle a **statistic**.

Statistics:

- **Sample Mean** - $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- **Sample Variance** - $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$

The realization of the above statistics are calculated by plugging in the actual values and are represented by lowercase letters. We divide by $n-1$ for variance to be correct across many samples.

Sampling distribution - the probability distribution of a statistic.

Key Results:

- For random samples of size n taken from an infinite population with mean μ_X and variance σ_X^2, the sampling distribution of the sample mean $\bar{X}$ has mean μ_X and variance $\frac{\sigma_X^2}{n}$
  - $\mu_{\bar{X}} = E(\bar{X}) = \mu_X$
  - $\sigma_{\bar{X}}^2 = V(\bar{X}) = \frac{\sigma_X^2}{n}$
- **Standard Error** - standard deviation of a sampling distribution is denoted by $\sigma_{\bar{X}}$. This describes how much $\bar{x}$ varies from sample to sample.
- **Law of Large Numbers** - $P(|\bar{X} - \mu| > \epsilon) \to 0$ as $n \to \infty$. Sample mean converges to population mean as n grows indefinitely.

5.3 Central Limit Theorem

If $\bar{X}$ is the mean of a random sample of size n taken from a population having mean μ and finite variance σ^2 , then, as $n \to \infty$,

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \to Z \sim N(0,1)$$

Or,

$$\overline{X} \to N\left(\mu, \frac{\sigma^2}{n}\right)$$

This means that the distribution of the sample mean tends towards a normal distribution as n gets larger. If $X_1, X_2, ...X_n$ follow a normal distribution, then $\overline{X}$ follows the normal distribution **exactly**.

5.4 Other Sampling Distributions - χ^2, t, F

5.4.1 The χ^2 Distribution

Let Z be a standard normal random variable. A random variable with the same distribution as Z^2 is called a χ^2 random variable with **one** degree of freedom.

Then, if we let $Z_1, ..., Z_n$ be independent and identically distributed standard normal random variables, a random variable with the same distribution as $Z_1^2 + \cdots + Z_n^2$ is called a χ^2 random variable with n degrees of freedom, denoted by $\chi^2(n)$.

Properties

1. Let $Y \sim \chi^2(n)$. $E(Y) = n$, $V(Y) = 2n$
2. For large n, $\chi^2(n) \sim N(n, 2n)$
3. If Y_1 and Y_2 are independent with m and n degrees of freedom respectively, $Y_1 + Y_2$ is a χ^2 random variable with $m+n$ degrees of freedom.
4. This distribution is a family of curves that are determined by the degrees of freedom, and all have a long right tail.

Upper Quartile for χ^2 We define $\chi^2(n;\alpha)$ such that for $Y \sim \chi^2(n)$, $P(Y > \chi^2(n;\alpha)) = \alpha$. Sort of like the upper quartile concept.

Considering the sample distribution of the random variable $\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{\sigma^2}$, it has a χ^2 distribution with $n-1$ degrees of freedom.

5.4.2 The t-Distribution

Suppose $Z \sim N(0,1)$ and $U \sim \chi^2(n)$. If Z and U are independent, then

$$T = \frac{Z}{\sqrt{U/n}}$$

follows the **t-distribution with n degrees of freedom.**

Properties

- Approaches $N(0,1)$ as $n \to \infty$, we can replace t-distribution when $n \geq 30$.
- If $T \sim t(n)$, then $E(T) = 0$ and $V(T) = \frac{n}{n-2}$ for $n > 2$
- The graph is symmetric about the vertical axis.

Upper Quartile for t-Distribution $t_{n;\alpha}$ such that for $T \sim t(n)$, $P(T > t_{n;\alpha}) = \alpha$

Connection between a random sample and the t-distribution If $X_1, \ldots, X_n$ are i.i.d. normal random variables with mean μ and variance σ^2 , then

$$\frac{\overline{X} - \mu}{S/\sqrt{n}}$$

follows a t-distribution with $n-1$ degrees of freedom.

5.4.3 The F-Distribution

Suppose $U \sim \chi^2(m)$ and $V \sim \chi^2(n)$. Then,

$$F = \frac{U/m}{V/n}$$

is called a **F-distribution** with (m,n) degrees of freedom, or $F(m,n)$.

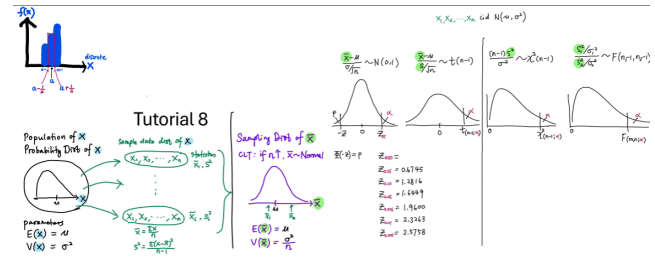
If $X \sim F(m,n)$ then $E(X) = \frac{n}{n-2}$ for $n > 2$,

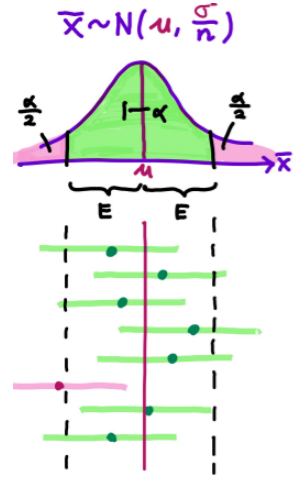
$V(X) = \frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$ for $n > 4$.

If $F \sim F(n,m), 1/F \sim F(m,n)$.

$P(F > F(m,n;\alpha)) = \alpha$.

We can show that $F(m,n;1-\alpha) = 1/F(n,m;\alpha)$.





Let θ be the parameter
$\hat{\theta}$ be the estimator

If $E(\hat{\theta}) = \theta$
$\hat{\theta}$ is an unbiased estimator of θ

$100(1-\alpha)\%$ CI for μ

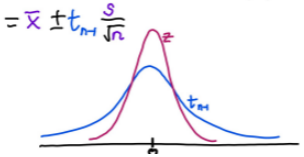
$= \overline{X} \pm E$

$= \overline{X} \pm z\frac{\sigma}{\sqrt{n}}$

$= \overline{X} \pm t_{n-1}\frac{s}{\sqrt{n}}$

$E = z\frac{\sigma}{\sqrt{n}}$

$n = \left(\frac{z\sigma}{E}\right)^2$



6 Estimation - Confidence Intervals

Two Types of Estimation - Point Estimation and Interval Estimation

6.1 Point Estimation

An **estimator** is a rule, usually expressed as a formula, that tells us how to calculate an estimate based on information in the sample.

Unbiased Estimator - if we let $\hat{\Theta}$ be an estimator of θ, if $E(\hat{\Theta}) = \theta$, we call $\hat{\Theta}$ an unbiased estimator of θ. Basically the mean value of it equals to the true value of the parameter.

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

is an unbiased estimator of σ^2 since $E(S^2) = \sigma^2$.

To check if a statistic is an unbiased estimator, find the expectation of it and check if the expected value = true value.

With probability $1-\alpha$, the error $|\overline{X} - \mu|$ is less than

$$E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Which is the **maximum error of estimate.** This is the variation from the middle of the curve where the value is acceptable.

Determination of Sample Size Minimum sample size so that with probability $1-\alpha$, the error is at most E_0. This implies $n \geq \left(\frac{z_{\alpha/2}\sigma}{E_0}\right)^2$. We need to quadruple the sample size to halve the error of estimate.

Case	Pop.	σ	n	Statistic	E	n for E_0, α
I	Normal	known	any	$Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$	$z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$	$\left(\frac{z_{\alpha/2}\cdot\sigma}{E_0}\right)^2$
II	any	known	large	$Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$	$z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$	$\left(\frac{z_{\alpha/2}\cdot\sigma}{E_0}\right)^2$
III	Normal	unknown	small	$T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$	$t_{n-1;\alpha/2} \cdot \frac{s}{\sqrt{n}}$	$\left(\frac{t_{n-1;\alpha/2}\cdot s}{E_0}\right)^2$
IV	any	unknown	large	$Z = \frac{\bar{X}-\mu}{S/\sqrt{n}}$	$z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$	$\left(\frac{z_{\alpha/2}\cdot s}{E_0}\right)^2$

6.2 Confidence Intervals for the Mean

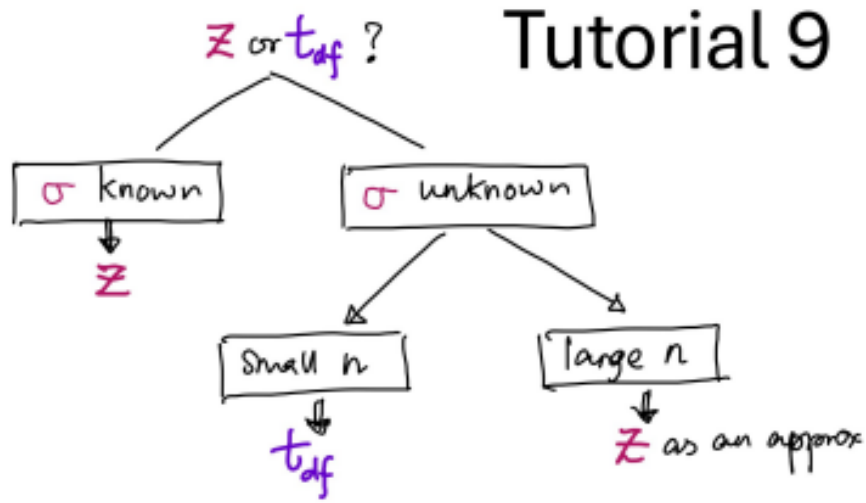
Interval estimator to calculate an interval (a, b) in which you are fairly certain the parameter of interest lies in. **Confidence level** is $(1-\alpha)$.

Confidence interval can be written as $\overline{X} \pm E$

Case	Population	σ	n	Confidence Interval
I	Normal	known	any	$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
II	any	known	large	$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
III	Normal	unknown	small	$\bar{x} \pm t_{n-1;\alpha/2} \cdot \frac{s}{\sqrt{n}}$
IV	any	unknown	large	$\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$

Note: n is considered large when $n \geq 30$.

Every time we construct a confidence interval, μ might be inside or it might not. We cannot say there's a chance it'll be inside, rather that if we construct this many times, about $(1-\alpha)$ of the confidence intervals will contain the true parameter. For example, if we construct a 95% confidence interval, 95 out of 100 confidence intervals will contain the true parameter.



6.3 Comparing Two Populations

We either have independent samples with complete randomization, or matched pairs samples, with randomization between matched pairs.

6.4 Independent Samples: Unequal Variances

Case 1: Two independent samples with **known** and **unequal** variances. Assuming either both populations are normal or both samples are large.
The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is then given as

$$(\overline{x} - \overline{y}) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Case 2: Two independent samples with **unknown** and **unequal** variances. Both samples are **large**.
The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given as

$$(\bar{x} - \bar{y}) \pm z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

6.5 Independent Samples: Equal Variances

Case 3: Two independent samples with **unknown** and **equal** variances. Both samples are **small**, and **normally distributed**.
We can use the **pooled estimator**, S_p^2

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$$

with S_1^2 and S_2^2 being the sample variances of the first and second samples respectively.

Then, we estimate σ^2 using S_p^2:

$$T = \frac{(\overline{X} - \overline{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

This follows a t-distribution with $n_1 + n_2 - 2$ degrees of freedom.
The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given as

$$(\overline{X} - \overline{Y}) \pm t_{n_1+n_2-2;\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Case 4: Two independent samples with **unknown** and **equal** variances, with both samples being **large**. We can replace the t-distribution by $z_{\alpha/2}$.
The $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given as

$$(\overline{X} - \overline{Y}) \pm z_{\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

We can roughly assume equal variance if $\frac{1}{2} \leq S_1/S_2 \leq 2$.

6.6 Paired Data

(X_i, Y_i) are matched pairs which are dependent on each other but independent with regard to other pairs. Define $D_i = X_i - Y_i,\ \mu_D = \mu_1 - \mu_2$. We can treat $D_1, D_2, ..., D_n$ as a random sample.
If n is small, and population is normally distributed, a confidence interval is

$$\bar{d} \pm t_{n-1;\alpha/2} \cdot \frac{s_D}{\sqrt{n}}$$

If n is large,

$$\bar{d} \pm z_{\alpha/2} \cdot \frac{s_d}{\sqrt{n}}$$

7 Hypothesis Testing - p-value, T-test

How to conduct a hypothesis test?

1. Set null and alternative hypothesis
   (a) Null hypothesis - default assumption. Usually $\mu =$ some value k
   (b) One-sided test: $H_1 : \mu > k$
   (c) Two-sided test: $H_1 : \mu \neq k$
2. Set level of significance
   (a) Reject H_0 when H_0 is true - **Type I error**
   (b) Not rejecting H_0 when H_0 is false - **Type II error**
   (c) **Level of Significance** - probability of making a type 1 error
   (d) Define $1 - \beta = P(\text{Reject } H_0 | H_0 \text{ is false})$ to be the power of the test.
3. Identify test statistic and distribution
   (a) To quantify how unlikely it is to observe the sample, **assuming the null hypothesis is true**
   (b) Dividing the possible values of the test statistic into two regions, rejection region and acceptance region.
4. Compute test statistic value
   (a) If value falls in rejection region, reject H_0 . If not, fail to reject.
5. Conclude, reject or fail to reject null hypothesis.

The p-value for the one-sided test ($H1 : > 0$) is half of that for the two-sided test ($H1 : \neq 0$).

7.1 Hypotheses Concerning the Mean

7.1.1 Case: Known Variance

Underlying distribution is normal OR n is sufficiently large (≥ 30).
Test Statistic:

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

For alternative hypothesis, if $H_1 : \mu \neq \mu_0$, rejection region is $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$.
For one sided, rejection region is $z > z_a$ or vice versa.
p-value - the probability of obtaining a test statistic at least as extreme than the observed sample value, given H_0 is true.
Then, if p-value $< \alpha$, reject H_0.

7.1.2 Case: Unknown Variance

Underlying distribution is normal. Test statistic given by:

$$T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

Same principle for rejection region based on two-sided vs one-sided. When $n \geq 30$, we can replace t by Z.

7.2 Two-Sided Tests and Confidence Intervals

Confidence intervals can be used to perform two-sided tests. If the interval does not contain the intended value, we can reject the null hypothesis.

7.3 Tests Comparing Means: Independent Samples

Drawing samples from 2 populations. Where $H_0 : \mu_1 - \mu_2 = \delta_0$
Case 1: Population variances are known and (underlying distributions are normal OR sample sizes are sufficiently large)

$$Z = \frac{(\bar{X} - \bar{Y}) - \delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim \mathcal{N}(0,1)$$

Case 2: Population variances are unknown and sample sizes are sufficiently large (≥ 30).

$$Z = \frac{(\bar{X} - \bar{Y}) - \delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim \mathcal{N}(0,1)$$

Case 3: Population variances are unknown but equal, underlying distributions are normal, and sample sizes are small.

$$Z = \frac{(\bar{X} - \bar{Y}) - \delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

Rejection Regions and p-values
For the null hypothesis $H_0 : \mu_1 - \mu_2 = \delta_0$, and specified alternative H_1, the rejection regions and p-values are given below.

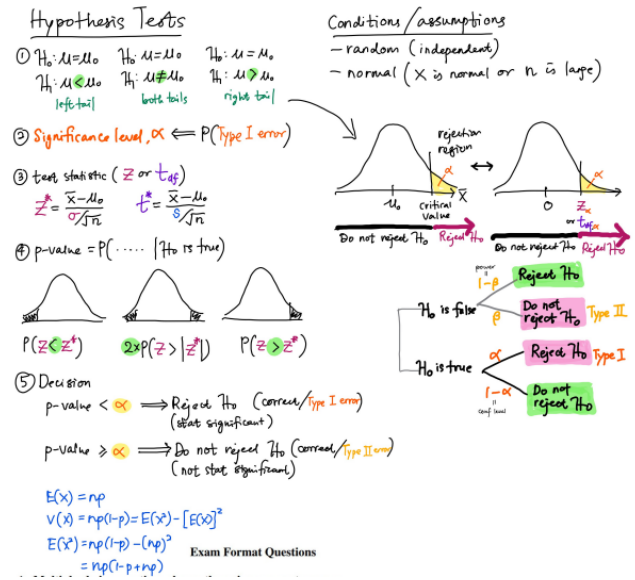
H_1	Rejection Region	p-value
$\mu_1 - \mu_2 > \delta_0$	$z > z_\alpha$	$P(Z > \lvert z \rvert)$
$\mu_1 - \mu_2 < \delta_0$	$z < -z_\alpha$	$P(Z < -\lvert z \rvert)$
$\mu_1 - \mu_2 \neq \delta_0$	$z > z_{\alpha/2}$ or $z < -z_{\alpha/2}$	$2P(Z > \lvert z \rvert)$

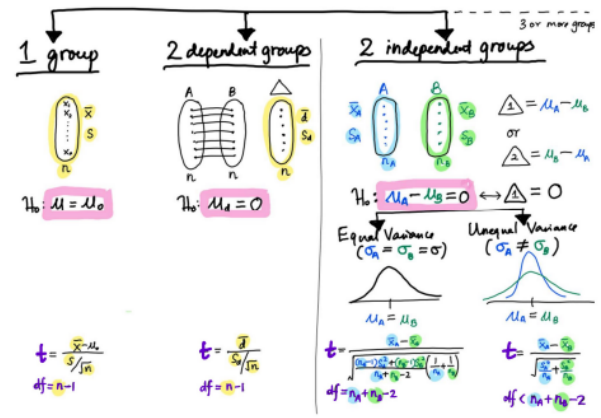
7.4 Tests Comparing Means: Paired Data

Define $D_i = X_i - Y_i$. Test statistic is given by:

$$T = \frac{\overline{D} - \mu_{D_0}}{S_D/\sqrt{n}}$$

If $n < 30$, $T \sim t_{n-1}$. Else, $T \sim N(0,1)$.





References:

- Wong Yean Ling's Tutorial Slides